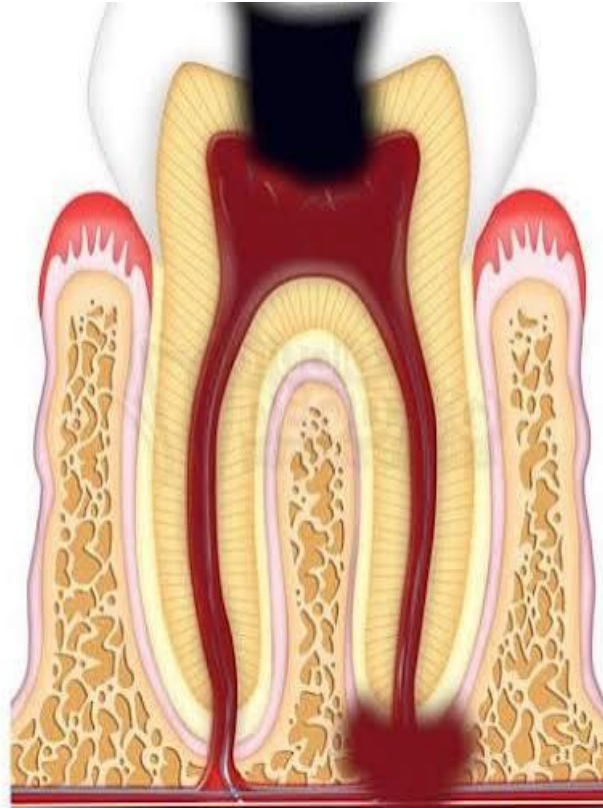




Diseases of Periapical Tissues

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Diseases of Periapical Tissues

The periapical tissue is composed of connective tissue, blood vessels, nerves, periodontal ligaments and the alveolar bone that collectively support and anchor the tooth.

Periapical tissue diseases are inflammatory conditions that affect the periapical tissue and surrounding structures, particularly around the apical portion of the root.

- Inflammation in the periapical part of the periodontal ligament is similar to that occurring elsewhere in the body. A particular feature of inflammation at this site is that the adjacent bone and occasionally the root apex may resorb. The periapical vascular network has a rich collateral circulation, which greatly enhances the ability of tissue to heal if the cause of inflammation is removed.
- Other feature is that the patients are often able to locate the symptoms to a specific tooth due to stimulation of the proprioceptive nerve endings in the periodontal ligament.
- In periapical periodontitis the pain is induced by pressure, percussion, or biting.

Etiology of periapical diseases:

- ✓ 1. Spread of bacterial infection following pulp necrosis.
- ✓ 2. Occlusal trauma.
- ✓ 3. Inadvertent endodontic procedures.
- ✓ 4. Infection through the gingival crevice.

Periapical diseases include

1. Acute periapical periodontitis.
2. Chronic periapical periodontitis (Periapical granuloma).
3. Periapical cyst (Apical periodontal cyst, Radicular cyst).
4. Periapical abscess (Dentoalveolar abscess).

Periapical Periodontitis(Apical periodontitis)

Definition:

It is an inflammation of periodontal tissues around apex of a non-vital tooth.

Types:

1. Acute periapical periodontitis.
2. Chronic periapical periodontitis (commonly presenting as a periapical granuloma).

Complications / Sequelae of periapical periodontitis

The consequences of periapical periodontitis depend on **the virulence of pathogens, the host response, and the effectiveness of any dental treatment received. If untreated, periapical periodontitis may progress to:**

1. Periapical granuloma (chronic inflammatory tissue at the apex)
2. Chronic periapical abscess.
3. Periapical(radicular) cyst.
4. Acute exacerbation of chronic periapical periodontitis (sudden flare-up of symptoms, leading to acute periapical abscess).
5. Spreading of acute periapical abscess into adjacent tissues, resulting in:
 - a. Osteomyelitis of the jaws (if the infection spreads into bone marrow spaces)
 - b. Cellulitis
 - c. Fascial space infections
 - d. Extraoral sinus tract formation (in severe chronic cases).

Acute Apical Periodontitis

It is acute periapical inflammation in periapical part of periodontal ligament.

Clinical features:

- Slight extrusion of the tooth from the socket.
- Tooth is tender on percussion due to inflammatory edema collected in the periodontal ligament.
- Most important feature is severe pain on slight pressure during mastication; due to external pressure, forcing of edema fluid against already sensitized nerve endings results in severe pain.
- Thermal changes do not induce pain.

Radiological features:

The radiographic appearances are often normal as there is generally insufficient time for bone resorption to occur. If radiographic changes are present, they consist of **slight widening of the periodontal ligament** and the lamina dura around the apex.

Histologic features:

1. Vascular dilatation.
2. Inflammatory cell infiltrate, predominantly neutrophils (PMNs).
3. Oedema in the periodontal ligaments situated in the confined space between the root apex and the alveolar bone, pus is not necessarily present.

Periapical Granuloma

Periapical granuloma is a localized mass of chronically inflamed granulation tissue at the apical region of non-vital tooth.

It is the most common periapical lesion.

Etiology\ pathogenesis:

- Usually develops in response to persistent low-grade infection from a necrotic\ infected pulp.
- Microbial product and inflammatory mediators exit through the apical foramen. This produces chronic inflammation in the periapical tissues.
- Osteoclast activation causes periapical bone resorption.
- The resorbed bone is replaced by vascular granulation tissue containing chronic inflammatory cells, forming a peri apical granuloma.
- It may also develop following an inadequately treated\ drained acute apical abscess.

Clinical features:

- Usually asymptomatic (most common)
- Sometimes a history of mild, dull, intermittent pain or slight discomfort during biting or chewing on solid foods.
- The patient may feel that the tooth is slightly extruded from the socket.
- May be slightly tender to percussion.
- No response to pulp vitality tests (due to pulp necrosis).
- May be discovered incidentally on radiograph during routine check-up.

Radiological features:

The periapical granulomas are usually **well-defined radiolucency**, but there are some situations when the lesions can appear ill-defined as in early stages, during acute exacerbation (acute abscess), or when healing\ inflammatory response is poor like in immunosuppression.

Histopathological Features:

- 1- Fibrous connective tissue capsule firmly attached to cementum.
- 2- Granulation tissue consists of newly formed blood vessels, fibroblasts, and collagen fibers.
- 3- Chronic inflammatory cells (lymphocytes, macrophages, and plasma cells).
- 4- Cholesterol clefts (slit like spaces) because the xylol used in the preparation of a histologic section dissolves the cholesterol crystals, leaving empty spaces (cholesterol clefts).
- 5- Russell bodies (immunoglobulin inclusions in plasma cells) may see.
- 6- Macrophages may accumulate to phagocyte the cholesterol clefts and called foam cells (foamy macrophages).
- 7- Foam cells might be found, which represent macrophages that contain lipoid material.
- 8- Multi nucleated giant cells could be seen around cholesterol clefts.
- 9- Epithelial cell rest of Malassez can be seen. They play a role in changing granuloma into cyst later on.

May also contain:

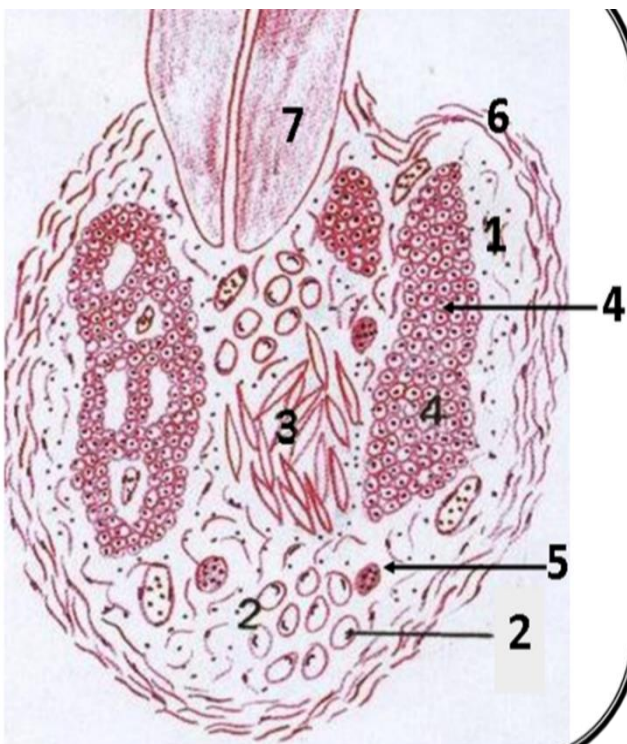
- ✓ Oral epithelium migrating from the periodontal pocket.
- ✓ Respiratory epithelium if the lesion is near the maxillary sinus.

Treatment:

- 1 -Tooth extraction.
- 2- Root canal therapy (RCT).
- 3- Endodontic treatment with apicectomy and curettage of the granuloma.

Sequalae of periapical granuloma:

- 1- Periapical granuloma may continue to enlarge with continued bone resorption.
- 2- Acute exacerbation to an acute periapical periodontitis.
- 3- A suppuration to form an acute periapical abscess.
- 4- Formation of a radicular cyst.
- 5- Low grade irritation may cause osteosclerosis (bone apposition) or cementum apposition (hypercementosis).



1. Granulation tissue.
2. Foam cells.
3. Cholesterol clefts.
4. Epithelial rests of Malassez.
5. Foreign body giant cells.
6. Fibrous CT capsule.
7. Apical root.

Periapical abscess

A periapical abscess is a localized collection of pus at the apex of a tooth root, resulting from bacterial infection of the pulp and periapical tissues. It may present as acute or chronic, with distinct clinical and radiographic features.

Acute Periapical Abscess

It is a localized suppurative inflammation (collection of pus) in the periapical area at the root apex of non-vital tooth.

Synonyms: Acute abscess, Acute dentoalveolar abscess, Acute periapical abscess, Acute radicular abscess.

Etiology

1. Extension of infection from pulp tissue.
2. May follow faulty endodontic treatment.
3. May result from acute exacerbation of periapical granuloma.
4. The causative organisms are pyogenic bacteria.

Clinical features:

- 1) Severe throbbing Pain, tooth is painful even to touch.
- 2) Tooth is usually extruded from its socket and mobile.
- 3) Localized swelling & erythematous change in overlying mucosa.
- 4) Tenderness on percussion and palpation.
- 5) Pain increases with systemic manifestations (fever, malaise & Lymphadenopathy).
- 6) No response to the pulp tester (non-vital tooth)
- 7) Pus discharge sometimes through sinus tract if chronicity develops.

Radiographic Feature:

Hazy or ill-defined radiolucent area around the apex of tooth.

Histopathological features:

1. Central area of liquefactive necrosis with pus (PMNLs, necrotic debris, bacteria)
2. Peripheral acute inflammatory infiltrate (PMNLs) with dilated vessels and edema.
3. Early tissue destruction and bone resorption at the apex.

Treatment:

- Systemic antibiotics if the periapical abscess is severe or has already spread to other parts of the body such as the neck to help bring the infection under control.
- Draining of pus through the root canal followed by endodontic treatment if the tooth is to be preserved.
- Extraction of the tooth.

Phoenix abscess is an acute exacerbation of a pre-existing chronic periapical lesion, usually a periapical granuloma, resulting in acute suppurative inflammation and abscess formation.

➤ Complication of periapical abscess:

- 1) Facial Cellulitis
- 2) Ludwig's angina
- 3) Osteomyelitis
- 4) Septicemia
- 5) Meningitis, brain abscess, cavernous sinus thrombosis

Facial cellulitis

- Facial cellulitis is an acute, diffuse, non-circumscribed bacterial infection of the skin and subcutaneous tissues of the face, characterized by erythema, warmth, swelling, and tenderness. Usually associated with malaise and fever.
- If untreated, it may lead to systemic involvement or severe complications, including orbital or intracranial spread.
- Cellulitis is more frequent in immunocompromised patients as diabetes mellitus, organ transplant and AIDS, malnutrition and alcoholism
- Infection spreads most commonly from mandibular third molars, whose apices are closely related to several facial spaces.
- Cellulitis arising from dental infection and spreading through soft tissues of head and neck can take various forms.
- Mostly, infection spreads through tissue spaces like canine space, infratemporal space, pharyngeal space, buccal space, submental and submandibular space etc.

Two especially dangerous forms of cellulitis

1. Cellulitis associated with mandibular teeth into submandibular and cervical tissues may cause (**Ludwig's angina**).
2. Cellulitis associated with maxillary teeth towards the eye may cause (**Cavernous sinus thrombosis**)

Clinical features:

- 1-Diffuse swelling, pain, fever, and malaise.
- 2- The swelling is tense and tender, often described as 'brawny' meaning swollen and hard, with a characteristic broad-like firmness.
- 3-The overlying skin is taut, shiny and edematous.
- 4-Limit opening of the mouth and often cause dysphagia.
- 5-Systemic manifestations include, fever, toxemia, leukocytosis and raised serum C-reactive protein (CRP) level.

6-Regional lymphadenitis.

➤ **Treatment:**

1-Immediate aggressive antibiotic treatment to prevent further spread of infection.

2-Remove the cause of infection.

3-Surgical drainage.

Ludwig's angina

➤ Ludwig's angina is a severe cellulitis involving submandibular, sublingual, and submental spaces bilaterally, resulting in life-threatening airway obstruction.

➤ In 70% cases develops from spread of infection from mandibular teeth.

➤ Increased prevalence in immunocompromised patients like AIDS, aplastic anemia, organ transplantation etc.

Clinical features:

1) Bilateral Submandibular Swelling: There is swelling on both sides under the chin, described as firm or brawny (indurated).

2) Odynophagia, drooling and trismus.

3) Tongue displacement(elevation) woody tongue.

4) Difficulty eating, swallowing(dysphagia), and breathing(dyspnea).

5) High fever, rapid pulse, and fast respiration.

6) Extension of inflammation to pharynx, larynx and suffocation and death.

Treatment:

1. Maintenance of the airway, tracheostomy may be indicated.

2. Antibiotic therapy.

3. Surgical drainage.

Cavernous sinus thrombosis (CST)

It is a rare, life-threatening blood clot that forms in the cavernous sinuses, which are located at the base of the brain, behind the eyes. It typically results from a bacterial infection that spreads from the face, teeth, ears, nose, or sinuses.

The clot can block blood flow, increase pressure and potentially damage the brain, eyes, and surrounding nerves, and if left untreated, can lead to serious complications like sepsis or even death.

Clinical features:

1. Patients may experience symptoms such as:
2. Severe headache, eye pain and fever.
3. Double vision.
4. Periorbital edema, proptosis (bulging eyes).
5. Chemosis (swelling of the conjunctiva).
6. Ophthalmoplegia (paralysis of eye movements) due to the pressure on cranial nerves.

Treatment:

1. High dose of penicillin.
2. Extraction and drainage (if fluctuant).
3. Corticosteroid and anticoagulant to prevent thrombosis and septic emboli formation.